Networks: Costs

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Practice quiz

You want to travel from Lausanne to Rome and you can choose among the following alternatives.

1. You can take the train from Lausanne to Geneva, which costs 20 CHF and has a travel time of 45 minutes. Then, you fly from Geneva to Rome, which costs 210CHF and has a travel time of 1h30.

2. You can use your car for a total cost of 150 CHF. However, since you have to be concentrated during 9h30 of the journey, it is estimated that each hour spent in the car has an additional cost of 5 CHF.

3. You can travel through Milan with a bus or a train. The bus costs 20CHF and takes 5h, while the train costs 60 CHF and takes 3h30. Once in Milan, you can reach Rome either by bus (40 CHF, 7h) or by train (90 CHF,3h).

We assume that you have a value of time of 30 CHF /hour. It means that each hour of travel is associated with a cost of 30CHF.

Model the resulting problem with a network representation. Mention the cost (in CHF) of each arc. The network must have an injective incidence function, meaning that there must be at most one arc between two nodes.